

Historical Transportability of Private-Capital SDFs

Which historical cash flows are relevant for pricing current funds?

July 15, 2026

Abstract

Private-capital SDF estimation faces a three-way data constraint. There are few vintage years, older vintages may transport poorly to current funds, and the most recent vintages have not yet produced the realized cash flows needed for appraisal-free estimation. We propose a relevance-weighted hybrid estimator that treats the historical information set itself as a model-selection object. Seasoned vintages contribute realized-cash-flow NPV moments; unseasoned vintages can contribute short-horizon NAV Euler errors; and historical vintages are weighted by temporal proximity and similarity in credit spreads, interest rates, and public-market valuations. Expanding, rolling, decaying, and state-conditioned SDFs are compared on a common NAV-free criterion. An ex-post transportability matrix measures how well an SDF estimated in one historical window prices earlier and later realized vintages. A forward pseudo-real-time exercise then jointly selects the SDF specification, the historical weighting scheme, and whether unseasoned Euler moments improve subsequent realized-cash-flow pricing relative to excluding young funds. The paper does not claim that thirty vintage years can cleanly distinguish a stable law from every form of temporal instability. It asks the empirically sharper question: which use of the available history travels best out of sample?

1 Introduction

How much historical private-capital data should be used to estimate an SDF for a fund raised today? Under a stable and sufficiently mixing data-generating process, older vintages add information and a full-history estimator becomes increasingly precise. Under temporal instability, however, the same estimator may converge precisely to a historical mixture that is no longer the relevant pricing law for current funds. More funds within a vintage can reveal that vintage's pricing moment more accurately, but they cannot create additional independent economic histories.

This creates a three-way empirical constraint. First, the time-series dimension is intrinsically thin: even a long private-capital database contains only a few dozen vintage years, and adjacent vintages live through overlapping calendar histories. Second, increasing the sample by reaching further into the past requires a transport assumption: that pricing relations estimated on older funds remain relevant for funds raised under different financing, valuation, and exit conditions. Third, restricting estimation to fully realized cash flows removes precisely the recent vintages that are most informative about the current environment. Conventional resampling can reduce fund noise inside the observed sample, but it cannot create economic time or resolve this maturity delay.

The private-equity pricing literature has developed several ways to recover risk and performance from irregular, non-traded cash flows. Driessen et al. [2012] estimate factor exposures and abnormal performance directly from fund cash flows, while Ang et al. [2018] recover a time series of private-equity returns from limited-partner cash flows. Korteweg and Nagel [2016] use a generalized public-market-equivalent approach to risk-adjust venture-capital cash flows. More recent work refines cash-flow benchmarking and SDF-based performance measurement [Gredil et al., 2023, Korteweg and Nagel, 2025]. These approaches differ in their maintained pricing restrictions and treatment of illiquidity, but they largely take the historical estimation sample as given. Our paper instead asks which part of that sample is informative for a current pricing problem: how well does an estimated private-capital SDF transport across economic time?

We call the empirical ability of an SDF estimated in one historical window to price cash flows from another window *historical transportability*. A stable pricing relation should travel across windows; a temporally unstable one should exhibit loss that changes with the distance and economic difference between estimation and evaluation periods. The concept does not require a literal regime classification. Smooth drift, isolated breaks, changes in the support of economic states, and changes in fund composition can all appear as failures of transportability. In an environment this short, a global test of “stability versus instability” is unlikely to have useful power against the many plausible alternatives. We therefore follow the forecasting-under-instability logic of evaluating local predictive performance rather than making a structural-break test the paper’s central result [Pesaran and Timmermann, 2007, Rossi, 2021].

Our proposed solution is a *forward-selected relevance-weighted hybrid SDF*. It has three components. First, candidate SDFs are estimated under a small, pre-specified sequence of historical weighting schemes: the full expanding sample, hard rolling windows, exponential time decay, and a time-and-state kernel. Second, recent economic similarity is measured with three observable state variables fixed before evaluating fund performance: credit spreads, the interest-rate level, and public-market valuations. Third, recent unseasoned vintages enter through short-horizon NAV Euler restrictions, but their inclusion is not maintained by assumption: the hybrid estimator is compared with an otherwise identical estimator that excludes them. Every choice is judged on later realized cash flows.

The empirical design has two parts. An ex-post transportability matrix uses fully realized vintages to show how pricing laws travel forward and backward across historical windows. Its backward entries are structural diagnostics, not feasible forecasts. A pseudo-real-time exercise then recreates historical information dates, estimates each candidate using the information available at that origin, and scores it on cash flows realized later by the then-young funds. This second exercise provides the directional evidence required to choose a current estimator.

The paper makes four contributions.

1. It formulates the private-capital data problem as a joint constraint of few economic histories, potentially decaying historical relevance, and delayed cash-flow outcomes.
2. It introduces historical transportability matrices and signed-distance loss curves that show where and how far a private-capital SDF travels.
3. It proposes a relevance-weighted SDF that combines simple temporal windows with low-dimensional economic-state similarity, while enforcing an effective-sample-size and identification constraint.
4. It evaluates the informational value of unseasoned NAV Euler moments on a common, later-realized cash-flow target rather than on NAV fit.

2 Methodology

2.1 The Private-Capital Data Trilemma

The empirical problem has three connected parts:

1. **Few vintages.** The relevant time-series sample contains only a few dozen vintage years, and overlapping fund lives make its effective temporal dimension smaller than its vintage count.
2. **Historical relevance.** Adding older vintages reduces sampling variance only if their pricing relations transport to the target period. Under temporal instability, a precisely estimated full-history SDF can be

precisely wrong for a current fund.

3. **Delayed outcomes.** The newest vintages contain the most current economic information but do not yet reveal their full realized cash flows. Discarding them mechanically makes a current estimator historical.

These are not three independent nuisances. Shortening the estimation window raises temporal relevance but increases sampling uncertainty and concentrates the sample in unseasoned funds. Requiring realized cash flows pushes the effective window backward. Reaching further into history restores moment count at the cost of a stronger transport assumption. The empirical objective is therefore not to eliminate any one component in isolation, but to find the combination that produces the lowest forward realized-cash-flow loss.

2.1.1 What the data can and cannot establish

Thirty vintage years do not support a clean omnibus decision between a time-invariant SDF and every form of temporal instability. A failure to reject stability may reflect low power, while a local episode of poor performance need not identify a permanent break. The paper therefore reports economically interpretable loss differences, transport curves, and model sets across a small pre-specified design space. Statistical tests are companions to this evidence, not the sole basis for declaring a regime change. In the terminology of the forecasting literature, local forecast performance is the target; parameter instability is one possible explanation for its variation [Rossi, 2021].

2.1.2 The cost of ignoring recent vintages

Discarding unseasoned vintages is clean only in a narrow measurement sense. It also changes the estimand by restricting the current SDF to funds raised far enough in the past to have substantially realized. Because buyout persistence and the composition of the industry have changed over time, this restriction cannot automatically be treated as innocuous [Harris et al., 2023]. The exclusion estimator is therefore an essential benchmark, not the default solution. Whether short-horizon Euler information improves forward pricing is an empirical question.

2.2 Pricing Moments

2.2.1 Seasoned vintages

For seasoned fund i of vintage v , let the realized pricing error per dollar committed be

$$g_i^S(\theta; M) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M) CF_{t,i},$$

and let $\epsilon_v^S(\theta; M)$ be its pooled vintage mean. The time-zero capital call enters undiscounted. The realized-cash-flow block is the common estimation and evaluation currency throughout the paper.

2.2.2 Unseasoned vintages

For an unseasoned fund, the full NPV moment is unavailable without valuing the residual claim. The baseline bridge uses the short-horizon NAV Euler error

$$e_{t,i}^H(\theta; M) = M_{t-1,t}(\theta; M) R_{t,i}^H - 1,$$

where $R_{t,i}^H$ is the quarterly Horizon-IRR return accounting for interim calls and distributions. Fund-quarter errors are aggregated and mapped into a vintage moment $\epsilon_v^U(\theta; M)$ using a pre-specified duration normalization. They are never discounted back to inception and never enter the comparison score. Reported NAVs are noisy appraisals, so this block is treated as a candidate information bridge rather than a traded-return restriction that must hold exactly. State-space corrected NAVs are a robustness extension, not the baseline [Brown et al., 2023].

The main unseasoned-treatment comparison is deliberately parsimonious:

1. *Exclude*: estimate only with seasoned realized-cash-flow moments;
2. *Hybrid*: add short-horizon NAV Euler moments for unseasoned vintages.

Both are evaluated on the identical subsequently realized cash flows. A more complicated measurement model earns a place in the main design only if the simple comparison demonstrates economically meaningful value from recent NAV information.

2.3 The Relevance-Weighted Hybrid SDF

At target date T , let z_v collect the economic state associated with vintage v . The baseline state vector contains three pre-performance variables: a credit spread, the interest-rate level, and a public-market valuation ratio. The variables are measured at or immediately before vintage formation and standardized using the estimation sample only.

For historical vintage v , define the unnormalized relevance weight

$$w_{v,T}(h, b) = \exp\left[-\log(2)\frac{T-v}{h}\right] \exp\left[-\frac{1}{2b^2}(z_v - z_T)'(z_v - z_T)\right],$$

where h is a time half-life and b is a common state-similarity bandwidth. The simple common bandwidth avoids estimating three separate state-sensitivity parameters from a few dozen vintages. Setting $h = \infty$ removes time decay; setting $b = \infty$ removes state conditioning. Hard rolling windows are obtained by replacing the time-decay component with an indicator.

Let q_v be a pre-specified reliability weight and normalize $w_{v,T}q_v$ to sum to one within the relevant estimation block. For candidate unseasoned treatment $A \in \{\text{Exclude, Hybrid}\}$, estimate

$$\hat{\theta}_T(M, h, b, A) = \arg \min_{\theta \in \Theta_M} \left\{ \sum_{v \in \mathcal{V}_T^S} \tilde{w}_{v,T} \epsilon_v^S(\theta; M)^2 + \mathbf{1}\{A = \text{Hybrid}\} \sum_{v \in \mathcal{V}_T^U} \tilde{w}_{v,T} \epsilon_v^U(\theta; M)^2 \right\}.$$

The design sequence moves from transparent to reasonably sophisticated while remaining feasible in a thin vintage panel:

1. **Expanding**: equal weight on the full available history;
2. **Rolling**: equal weight inside a fixed recent-vintage window;
3. **Decay**: exponential time weights indexed by half-life h ;
4. **Time-state**: exponential decay combined with similarity in credit spreads, interest rates, and public valuations.

A regularized time-varying parameter path of the form

$$\min_{\{\theta_v\}} \sum_v \epsilon_v(\theta_v)^2 + \lambda \sum_{v>1} \|\theta_v - \theta_{v-1}\|_2^2$$

is the principal sophistication check [Cui et al., 2024]. It is not the headline estimator unless it produces a clear forward improvement over the lower-dimensional weighting schemes.

Every candidate must satisfy two feasibility restrictions before it can be scored: the weighted moment Jacobian must be sufficiently well conditioned, and the weight concentration must leave an effective vintage count

$$V_{\text{eff}}(T) = \frac{1}{\sum_v \tilde{w}_{v,T}^2}$$

comfortably above the number of SDF parameters. This prevents an apparently local estimator from being driven by one historically similar vintage.

2.4 Historical Transportability

2.4.1 The transportability matrix

Let s index a historical estimation window and t an excluded, fully realized evaluation vintage. Define the NAV-free transport loss

$$L_{s \rightarrow t}(M, h, b) = \hat{U}_t^{\text{real}} \left(\hat{\theta}_s(M, h, b, \text{Exclude}) \right),$$

where the evaluation score uses only realized fund cash flows. The exclusion version isolates historical transportability from the separate question of how young-fund NAV information enters. Stacking these losses over estimation windows and evaluation vintages yields a *historical transportability matrix*. Its rows are window SDFs, its columns are evaluation vintages, and each entry measures how well a pricing law travels across economic time.

Aggregating by signed temporal distance $d = t - s$ gives the transport loss curve

$$\mathcal{T}_{M,h}(d) = \frac{1}{N_d} \sum_s L_{s \rightarrow s+d}(M, h, b).$$

Positive distances measure forward transportability; negative distances measure backward transportability. Only the former has a real-time forecasting interpretation. Backward transportability is a diagnostic of structural stability, not a feasible investment rule.

The matrix is descriptive by design. A flat surface is consistent with broad historical transportability; loss increasing in $|d|$ is consistent with temporal localization; and isolated high-loss columns indicate historically unusual target periods. None of these patterns alone proves a particular structural-break model.

2.4.2 Calendar-time overlap

Adjacent vintage moments share public-factor histories because their funds remain active for many years. We therefore measure pairwise calendar-time overlap rather than treating vintage labels as independent by construction. Let $a_{v,t}$ denote vintage v 's discounted-cash-flow or Jacobian exposure to calendar period t . Define

$$O_{vv'} = \frac{\sum_t a_{v,t} a_{v',t}}{\sqrt{\sum_t a_{v,t}^2} \sqrt{\sum_t a_{v',t}^2}}.$$

The overlap matrix is reported alongside the transportability matrix and is used to purge near-duplicate evaluation windows where feasible. Because complete separation would eliminate most of the data, overlap is quantified and stress-tested rather than treated as a binary admissibility standard.

2.5 Unseasoned Vintages as Delayed Outcomes

At historical information date τ , let A denote a bridge for unseasoned vintages and estimate

$$\hat{\theta}_{\tau,h,A}(M) = \arg \min_{\theta \in \Theta_M} Q_{\tau,h,A}(\theta; M)$$

using only fund information dated no later than τ . Candidate bridges are compared on the identical realized-cash-flow target after the then-unseasoned vintages have seasoned:

$$L_{\tau \rightarrow \tau+\ell}(M, h, A) = \frac{1}{|\mathcal{V}_{\tau}^U \cap \mathcal{V}_{\tau+\ell}^S|} \sum_{v \in \mathcal{V}_{\tau}^U \cap \mathcal{V}_{\tau+\ell}^S} \hat{U}_v^{\text{real}}(\hat{\theta}_{\tau,h,A}(M)).$$

The empirical comparison therefore selects a triplet: the SDF specification, the historical weighting rule, and the unseasoned-vintage treatment. Exclusion remains in the race as the benchmark measuring the cost of delayed outcomes; the short-horizon Euler block is the proposed simple bridge. The paper presents the configuration with the lowest forward loss as its preferred empirical solution, subject to the feasibility and stability diagnostics above.

2.6 Evaluation Design

Ex-post transportability. Estimate local SDFs on fully observed historical windows and score them on earlier and later seasoned vintages. This exercise is fully NAV-free and isolates temporal transportability, but it does not claim a real-time interpretation when later information is used to estimate an SDF that is scored on an earlier period.

Pseudo-real-time delayed-outcome validation. Reconstruct a sequence of historical information dates, estimate each window/bridge combination using the information fields available at the corresponding date, and score it on subsequently realized fund cash flows. A literal real-time reconstruction would also require historical database snapshots, contemporaneous coverage records, and knowledge of which observations were later backfilled. Imposing that ideal would leave essentially no usable evaluation sample. We therefore adopt the pragmatic pseudo-real-time standard feasible in the data and state its residual look-ahead limitation explicitly.¹

At each origin, tuning uses only earlier admissible origins. The outer origin is reserved for evaluation. With few origins, the tuning grid is intentionally coarse and common across SDF specifications. The empirical result is reported as a loss surface and a set of near-optimal configurations, alongside the single preferred configuration, so that small differences are not given a false winner-takes-all interpretation.

¹The fully reconstructed real-time ideal is conceptually useful even when empirically infeasible: it identifies the remaining source of look-ahead without pretending that an empty sample is a superior research design.

2.7 Forward Selection Criterion

Let O denote the admissible pseudo-real-time origins and $\mathcal{H}_\tau$ the vintages that were unseasoned at τ and later become eligible for realized-cash-flow scoring. For design $c = (M, h, b, A)$, define

$$\widehat{R}(c) = \frac{1}{|O|} \sum_{\tau \in O} \frac{1}{|\mathcal{H}_\tau|} \sum_{v \in \mathcal{H}_\tau} \widehat{U}_v^{\text{real}}(\widehat{\theta}_\tau(c)).$$

The preferred design minimizes $\widehat{R}(c)$. All designs face the same origins and realized evaluation funds, so comparisons are paired. The paper emphasizes effect sizes, origin-by-origin stability, and sensitivity to the coarse tuning grid. With only a few dozen vintage years, formal rejections of equal predictive loss are interpreted cautiously.

2.8 Inference and Reporting

Model comparisons use paired loss differences because every admissible design is evaluated on the same origin–vintage cells. Losses are first averaged within origin so that origins with many funds or newly seasoned vintages do not dominate mechanically. We report the mean paired difference, its distribution across origins, and block-bootstrap uncertainty that preserves adjacent-origin dependence. Parameter uncertainty for a selected SDF is assessed with vintage- level resampling and calendar-overlap blocks. Given the short time series and the additional uncertainty created by model selection, confidence intervals and predictive-loss tests are interpreted as measures of precision rather than as a binary stability verdict.

3 Data and Empirical Design

This section maps the empirical inputs directly into the moments, state weights, and forward loss defined above. The ordering is deliberate: we first construct an appraisal-free realized-cash-flow target, then define what was observable at each historical origin, and only then compare alternative uses of the available history.

3.1 Private-Capital Cash-Flow Data

The fund data are drawn from [TBD: data provider and extraction date]. The unit of observation is a dated limited-partner cash flow for a [TBD: buyout / broader private-capital] fund. Required fields are fund identifier, vintage year, commitment, dated capital calls, dated distributions, reported NAVs, strategy, geography, and — where available — general-partner identifier. Cash flows and NAVs are measured net of management fees and carried interest from the limited partner’s perspective and converted to [TBD: currency and real/nominal units]. We exclude funds with internally inconsistent cash-flow histories, missing commitments, nonpositive commitments, or implausible scale ratios under a pre-specified cleaning rule. The analysis retains all admissible funds within a vintage; it does not create pseudo-vintages by resampling funds.

The first data table should make the thin time-series dimension visible rather than reporting only a large fund count:

Table 1: Private-capital sample by vintage period

Vintage period	Vintages	Funds	Commitments	Funds/vintage	Realized share
[TBD: early period]					
[TBD: middle period]					
[TBD: recent period]					
Full sample					

Notes: Commitments are reported in [TBD: units]. Funds per vintage is the median across vintage years. Realized share is the fraction of funds satisfying the baseline seasoning rule at the final sample date.

3.2 Seasoning and the Realized Evaluation Sample

At information date τ , a fund is seasoned when its residual value share

$$s_{i,\tau} = \frac{\text{RVPI}_{i,\tau}}{\text{DPI}_{i,\tau} + \text{RVPI}_{i,\tau}}$$

is at most δ . The proposed baseline is $\delta = 0.15$, with 0.10 and 0.20 as pre-specified robustness thresholds. A fund that fails this rule may contribute to the hybrid short-horizon Euler block but never to the realized evaluation loss at that date. The realized target includes only calls and distributions observed by the evaluation date; the residual NAV is used to determine eligibility but is neither priced nor added as a terminal cash flow. Thus, δ trades bounded terminal truncation for sample size without fitting the score to NAV; $\delta = 0$ is the strict robustness check.

We report, for every pseudo-real-time origin, the number of available vintages, seasoned and unseasoned funds, the fraction of commitment represented by the realized evaluation set, and the later vintages that can eventually be scored. This origin-by-origin coverage table is essential: it reveals whether an apparent forecasting gain comes from a genuinely broader information set or from a small and unusual set of funds that happens to mature early.

3.3 Public Factors and Candidate SDFs

Public-market returns and the risk-free rate are obtained from [TBD: source] at the same frequency as the cash-flow discounting exercise. The main comparison uses a deliberately small SDF set: a risk-free benchmark, a one-factor public- equity SDF, a levered public-equity specification, and at most one established multifactor alternative. This keeps the empirical question focused on the historical information set rather than a factor zoo. Factor definitions, currency, return convention, and the mapping from quarterly factors to dated cash flows are fixed before inspecting forward pricing losses.

3.4 Economic-State Variables

The state vector z_v contains one variable for each of three economic dimensions: a corporate credit spread, the interest-rate level, and a public- equity valuation ratio. The proposed baseline definitions are the high-yield option-adjusted spread, a long-term nominal government yield, and the cyclically adjusted price–earnings ratio [TBD: or a price–dividend ratio]. To avoid using fund outcomes to define similarity, each variable is measured as its average over the 12 months ending immediately before the vintage year. At each origin, levels are standardized using only the available estimation vintages. We also report the historical support of z_T ; a target outside that support is an extrapolation problem and should not receive a falsely precise local estimate.

3.5 Pseudo-Real-Time Origins

Annual origins begin at the first date for which all candidate SDFs satisfy the minimum effective-vintage and Jacobian conditions. At each origin, cash flows, NAVs, factors, and state variables are truncated to their admissible information date. Hyperparameters are chosen using only earlier origins, and the current origin is used once for evaluation. The final extract does not reconstruct historical database coverage or later backfills; this pragmatic limitation is disclosed and tested by excluding observations with suspiciously late first reporting where such timestamps exist. The stronger literal real-time standard would otherwise reduce the usable evaluation sample to essentially zero.

3.6 Pre-Specified Design Grid

The grid is intentionally coarse. Its final endpoints are fixed after a descriptive feasibility audit, but before comparing out-of-sample losses.

Table 2: Pre-specified empirical design

Object	Baseline candidates	Restriction or robustness
SDF	Risk-free; public equity; levered equity; one multifactor model	Common factor set at every origin
Historical weighting	Expanding; rolling; exponential decay; time–state	Same candidate sequence for every SDF
Rolling length	8, 12, 16 vintages	Retained only when identification is adequate
Time half-life	4, 8, 12 vintage years	$h = \infty$ reproduces no decay
State bandwidth	0.75, 1.5, 3 standardized units	$b = \infty$ reproduces time-only weighting
Young-fund treatment	Exclude; short-horizon NAV Euler hybrid	Common later realized-cash-flow target
Seasoning threshold	Residual value share at most 15%	10%, 20%, and zero residual value
Feasibility	$V_{\text{eff}} \geq \max\{8, 3 \dim(\theta)\}$	Jacobian conditioning reported separately

3.7 Empirical Workflow

The empirical analysis proceeds in the same order as the methodology:

1. construct seasoned realized-cash-flow moments and document calendar-time overlap across vintages;
2. estimate fully realized local SDFs and form the ex-post historical transportability matrix;
3. reconstruct pseudo-real-time origins and calculate feasible expanding, rolling, decaying, and time–state estimates;
4. select tuning parameters using earlier origins only;
5. score all candidates on the same later realized-cash-flow outcomes and compare exclusion with the short-horizon NAV Euler hybrid; and
6. re-estimate the selected specification at the final information date and report both its current SDF parameters and the historical weights that support them.

4 Empirical Results

4.1 The Effective Historical Sample

Table 1 first documents the nominal and effective sample size. A vintage-by-calendar-year cash-flow heat map shows the long overlap of fund lives, while the overlap matrix O summarizes common factor exposure. We report the eigenvalue-based effective temporal rank and effective vintage counts under each candidate weighting rule. This section establishes how much independent economic history is actually available before asking whether old history remains relevant.

4.2 Ex-Post Historical Transportability

The first main figure is the historical transportability matrix for the baseline SDF. Rows identify estimation windows, columns identify excluded realized vintages, and colors represent realized pricing loss. Signed-distance loss curves then summarize forward and backward transport. The empirical questions are whether loss rises with temporal distance, whether the pattern is asymmetric, and whether particular evaluation vintages are difficult for every historical window.

4.3 How Much History Travels Forward?

The main model-comparison table reports paired forward loss for the expanding, rolling, decay, and time-state estimators on common origin-vintage cells. For each design we report average loss, the difference relative to expanding history, origin-level uncertainty, V_{eff} , and the fraction of origins passing identification checks. This table supplies the paper’s primary answer: whether using less, or selectively weighted, history improves pricing of later realized private-capital cash flows.

4.4 Does Economic-State Similarity Add Information?

We next hold the time-weighting rule fixed and compare time-only with time-state weights. Weight plots show which distant vintages are recovered because their credit spreads, interest rates, or public valuations resemble the target. Leave-one-state-variable-out results reveal whether improvement is broad-based or driven by a single state proxy. A state-conditioned estimator is preferred only if it improves forward loss without collapsing the effective vintage count.

4.5 Do Unseasoned Vintages Improve the Current SDF?

For every admissible origin, we compare the exclusion and short-horizon NAV Euler estimators using identical later realized cash flows. The main table reports paired loss differences overall and by the age of the young funds at the estimation origin. We also relate the hybrid’s gain to its NAV share and subsequent NAV revisions. This distinguishes useful early information from a mechanical fit to smoothed reported values.

4.6 The Preferred SDF and Current Risk-Adjusted Performance

The final results section re-estimates the forward-selected design at the current information date. We report SDF parameters, implied factor exposures, risk-adjusted performance, the weights assigned to every historical

vintage, and sensitivity to all configurations whose forward loss is statistically and economically close to the minimum. The conventional expanding-history estimate is shown alongside it so that the effect of historical transportability is transparent.

5 Robustness and Method Validation

5.1 Alternative Sample and Moment Choices

Robustness exercises vary the seasoning threshold, exclude residual NAV entirely, remove the financial-crisis and pandemic vintages one at a time, alter fund-size reliability weights, and restrict the sample by strategy or geography where coverage permits. Results are also reported with vintage- equal weighting so that a few data-rich vintage years cannot determine the conclusion.

5.2 Alternative Windows, States, and SDFs

We vary the coarse window grid, use a real rather than nominal long rate, replace the valuation proxy, winsorize state distances, and compare the common bandwidth with a strongly regularized variable-specific bandwidth. The smooth time-varying parameter path is retained as a sophistication check. Alternative SDFs matter only insofar as the transportability result survives reasonable pricing specifications.

5.3 Placebos

Two placebos discipline the interpretation of state similarity. First, state vectors are permuted across vintages while their time ordering is retained. Second, the target state is replaced by a future state that was unavailable at the origin. Genuine transport gains should disappear under the first placebo; the second quantifies how much an inadvertently forward-looking state construction could exaggerate performance.

5.4 Compact Method Validation

A standalone simulation section is not required for the paper’s main empirical claim. The estimator is selected by observed forward realized-cash-flow loss, and a large menu of simulated breaks would not establish which process generated the single historical private-capital sample. A compact appendix calibration can nevertheless provide a useful diagnostic. It should match the empirical vintage count, cash-flow-age profile, and calendar overlap under only two designs — a stable SDF and a smoothly drifting SDF — each with and without appraisal smoothing. Its limited purpose is to verify that the selection procedure does not mechanically prefer short windows, state conditioning, or the NAV bridge when the pricing law is stable. The empirical transport and delayed-outcome exercises remain the evidence of interest.

6 Conclusion

Private-capital SDF estimation cannot solve a thin and delayed economic history by resampling funds more aggressively. The relevant choice is which historical vintages to transport to the target period and whether partial information from recent funds improves that transport. The proposed estimator makes both choices empirical: it weights history by time and observable economic similarity, admits recent NAV information only through a short-horizon bridge, and selects the complete design by its ability to price later realized cash flows. The result is a current SDF supported by the part of history that has demonstrated the greatest forward relevance, rather than by the longest history available.

References

- Andrew Ang, Bingxu Chen, William N. Goetzmann, and Ludovic Phalippou. Estimating private equity returns from limited partner cash flows. *The Journal of Finance*, 73(4):1751–1783, 2018. doi: 10.1111/jofi.12688.
- Gregory W Brown, Eric Ghysels, and Oleg R Gredil. Nowcasting net asset values: The case of private equity. *The Review of Financial Studies*, 36(3):945–986, 2023.
- Liyuan Cui, Guanhao Feng, and Yongmiao Hong. Regularized GMM for time-varying models with applications to asset pricing. *International Economic Review*, 65(2):851–883, 2024. doi: 10.1111/iere.12678.
- Joost Driessen, Tse-Chun Lin, and Ludovic Phalippou. A new method to estimate risk and return of non-traded assets from cash flows: The case of private equity funds. *Journal of Financial and Quantitative Analysis*, 47(3):511–535, 2012. doi: 10.1017/S0022109012000221.
- Oleg R. Gredil, Barry Griffiths, and Rüdiger Stucke. Benchmarking private equity: The direct alpha method. *Journal of Corporate Finance*, 81:102360, 2023. doi: 10.1016/j.jcorpfin.2023.102360.
- Robert S. Harris, Tim Jenkinson, Steven N. Kaplan, and Ruediger Stucke. Has persistence persisted in private equity? evidence from buyout and venture capital funds. *Journal of Corporate Finance*, 81:102361, 2023. doi: 10.1016/j.jcorpfin.2023.102361.
- Arthur Korteweg and Stefan Nagel. Risk-adjusting the returns to venture capital. *The Journal of Finance*, 71(3):1437–1470, 2016. doi: 10.1111/jofi.12380.
- Arthur Korteweg and Stefan Nagel. Risk-adjusted returns of private equity funds: A new approach. *The Review of Financial Studies*, 38(9):2557–2601, 2025. doi: 10.1093/rfs/hhae067.
- M. Hashem Pesaran and Allan Timmermann. Selection of estimation window in the presence of breaks. *Journal of Econometrics*, 137(1):134–161, 2007. doi: 10.1016/j.jeconom.2006.03.010.
- Barbara Rossi. Forecasting in the presence of instabilities: How we know whether models predict well and how to improve them. *Journal of Economic Literature*, 59(4):1135–1190, 2021. doi: 10.1257/jel.20201479.